# logdist effect of a covariate

## Parametrization

This model implements a non-linear effect of a positive covariate $x$ as a part of the linear predictor,

$$\beta\,(1 + \exp(\alpha_1 \log(x) - \alpha_2 x)$$

where $\beta \in \Re$, $\alpha_1, \alpha_2 \in \Re^+$ and $x \geq 0$.

## Hyperparameters

This model has three hyperparameters, the scaling $\beta$, $\alpha_1$ and $\alpha_2$.

$$\theta_1 = \beta \qquad \theta_2 = \log(\alpha_1) \qquad \theta_3 = \log(\alpha_2)$$

and the priors are given for $\theta_1, \theta_2$ and $\theta_3$.

## Specification

```
f(x, model="logdist", hyper = ..., precision = <precision>)
```

where `precision` is the precision for the tiny noise used to implement this as a latent model.

### Hyperparameter specification and default values

**doc** `A nonlinear model of a covariate`

**hyper**

**theta1**

**hyperid** `39021`
**name** `beta`
**short.name** `b`
**initial** `1`
**fixed** `FALSE`
**prior** `normal`
**param** `0 1`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta2**

**hyperid** `39022`
**name** `alpha1`
**short.name** `a1`
**initial** `0`
**fixed** `FALSE`
**prior** `loggamma`
**param** `0.1 1`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta3**

**hyperid** 39023
**name** alpha2
**short.name** a2
**initial** 0
**fixed** FALSE
**prior** loggamma
**param** 0.1 1
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**constr** FALSE

**nrow.ncol** FALSE

**augmented** FALSE

**aug.factor** 1

**aug.constr**

**n.div.by**

**n.required** FALSE

**set.default.values** FALSE

**pdf** logdist

## Example

```
logdist = function(x, beta,  alpha)
{
    return (beta * (1 + exp(alpha[1] * log(x) - alpha[2] * x)))
}

n = 1000
s=0.1
x = runif(n)
beta = 1
alpha = c(1, 0.5)
## start at the true values
hyper = list(
    beta = list(initial = beta),
    a1 = list(initial = log(alpha[1])),
    a2 = list(initial = log(alpha[2])))
## start somewhere else
hyper = list(
    beta = list(initial = 1),
    a1 = list(initial = 0),
    a2 = list(initial = 0))

y = logdist(x, beta,  alpha) + rnorm(n, sd = s)
r = (inla(y ~ -1 + f(x, model="logdist", hyper = hyper),
```

```
          data = data.frame(y, x),
          family = "gaussian",
          verbose=TRUE,
          control.inla = list(h=0.0001),
          control.family = list(
              hyper = list(
                  prec = list(
                      initial = log(1/s^2),
                      fixed = TRUE)))))
summary(r)
```

### Notes

None